

Question	Answer	Marks
3(a)	$a = -\omega^2 x$	M1
	a = acceleration, x = displacement from equilibrium position and ω = angular frequency	A1
3(b)(i)	$x_0 = 0.12 \text{ m}$	A1
3(b)(ii)	$v = \omega\sqrt{(x_0^2 - x^2)}$ two (x, v) pairs correctly read from Fig. 3.2 (one may be $(x_0, 0)$ or value of x_0 from (i))	C1
	e.g. $0.20 = \omega\sqrt{(0.12^2 - 0)}$ leading to $\omega = 1.7 \text{ rad s}^{-1}$	A1
3(b)(iii)	$E = \frac{1}{2}M\omega^2 x_0^2$	C1
	$0.050 = \frac{1}{2} \times M \times 1.67^2 \times 0.12^2$ $M = 2.5 \text{ kg}$	A1
	or	
	$(E_K)_{\max} = \frac{1}{2}Mv_0^2$	(C1)
	$0.050 = \frac{1}{2} M \times 0.20^2$ $M = 2.5 \text{ kg}$	(A1)
3(c)(i)	loss of (total) energy (of system)	B1
	due to resistive forces	B1
3(c)(ii)	closed loop surrounding the origin with maximum x at $\pm 0.060 \text{ m}$ passing through $v = 0$	B1
	maximum velocity shown as $\pm 0.10 \text{ m s}^{-1}$ passing through $x = 0$	B1

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4(c)	(field line indicates) direction of force	D1